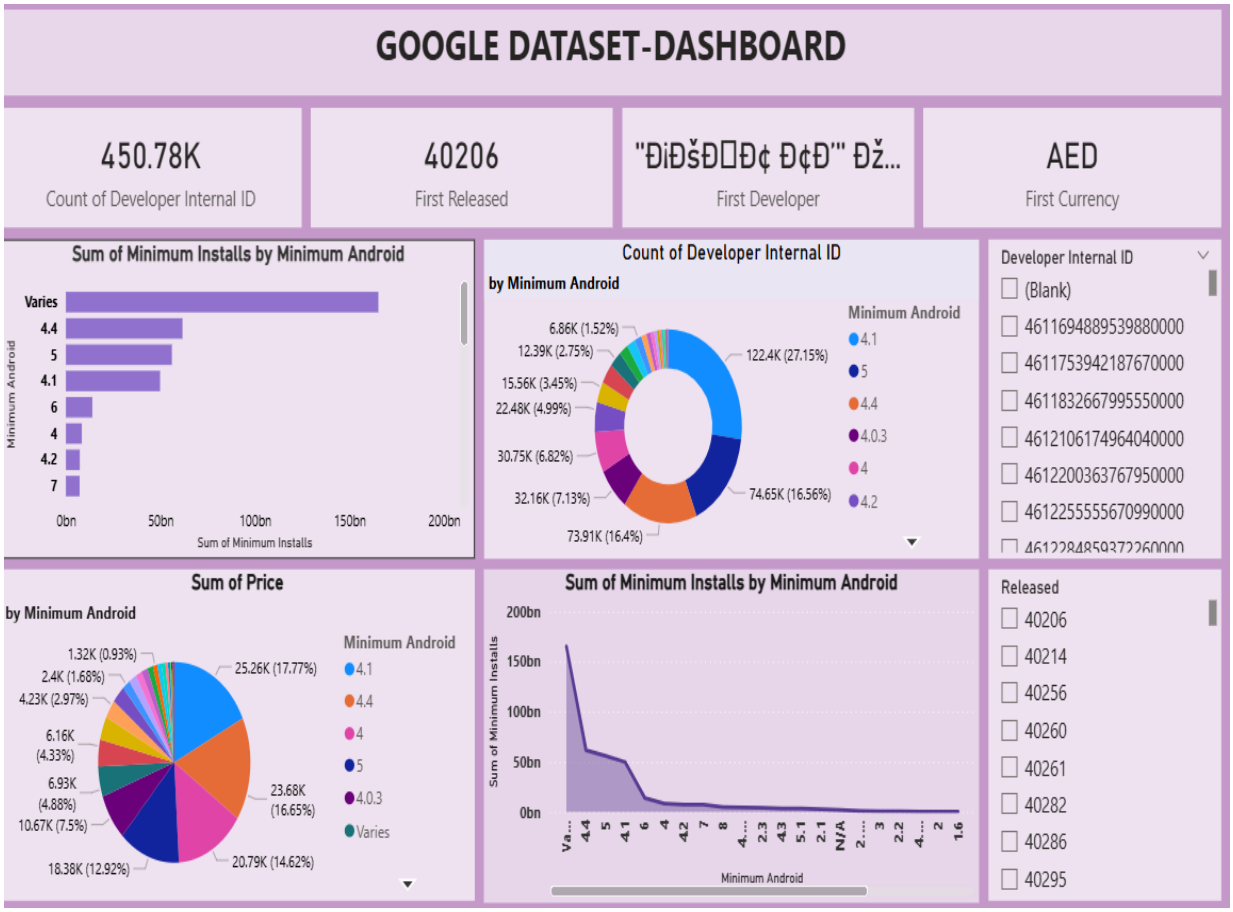
Google Play Store Data Analysis

1. Introduction

In today's digital world, mobile applications play a vital role in everyday life. The Google Play Store hosts millions of applications across various categories, making it essential to analyze app performance, user engagement, and market trends. This project focuses on analyzing a Google Play Store dataset using Power BI to generate meaningful insights. The dashboard provides a comprehensive visualization of app installs, pricing, Android version compatibility, and developer distribution.



2. Objective of the Project

The main objectives of this project are:

- To analyze Google Play Store app data effectively
- To understand app installation trends across Android versions
- To evaluate pricing strategies of apps
- To identify developer contributions and app distribution
- To create an interactive dashboard using Power BI

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3. Dataset Description

The dataset used in this project contains information about applications available on the Google Play Store. The key attributes include:

- Developer Internal ID
- App Release Date
- Minimum Android Version
- Number of Installs
- Price of the App
- Currency Type

The dataset consists of over **450K records**, providing a large-scale view of app performance and distribution.

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4. Tools and Technologies Used

- **Power BI** – Data visualization and dashboard creation
- **Microsoft Excel** – Data preprocessing and cleaning
- **Data Analysis Techniques** – Aggregation, filtering, and trend analysis

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5. Data Cleaning and Preparation

Before visualization, the dataset was cleaned and prepared:

- Removed null and duplicate values
- Standardized Android version formats
- Converted data types (e.g., price, installs)
- Filtered irrelevant or inconsistent records

This step ensured accurate and reliable analysis.

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6. Dashboard Overview

The Power BI dashboard provides key insights using multiple visualizations:

Key Metrics

- Total Developer IDs: **450.78K**
- First Released Record: **40206**
- First Developer Name: Displayed in dashboard
- Currency Used: **AED**

Visualizations Included

1. **Bar Chart – Minimum Installs by Android Version**
   Shows how installs vary across different Android versions.
2. **Donut Chart – Developer Distribution**
   Represents the count of developers by minimum Android requirement.
3. **Pie Chart – Sum of Price by Android Version**
   Highlights pricing trends among different Android versions.
4. **Line Chart – Install Trends**
   Displays how installs change across Android versions.
5. **Filters (Slicers)**
   - Developer Internal ID
   - Release Date

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7. Key Insights

- Apps supporting **lower Android versions** have significantly higher installs, indicating broader device compatibility.
- The majority of apps are concentrated around Android versions **4.1 to 5.0**.
- Pricing varies significantly, but most apps are either free or low-cost.
- A small number of developers contribute a large portion of applications.
- Install trends decrease as Android version requirements increase.

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8. Advantages of the Project

- Helps developers understand market demand

- Assists businesses in pricing strategy decisions
- Provides insights into user accessibility based on Android versions
- Enhances decision-making using data visualization

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9. Limitations

- Dataset may contain outdated or incomplete records
- Some fields (like developer name) may not be fully accurate
- Analysis is limited to available dataset attributes

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10. Future Enhancements

- Integration with real-time Play Store data
- Advanced analytics using machine learning
- Predictive modeling for app success
- Category-wise and user review analysis

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11. Conclusion

This project successfully demonstrates how data visualization tools like Power BI can transform raw data into actionable insights. By analyzing the Google Play Store dataset, we identified important trends in app installs, pricing, and Android compatibility. The dashboard enables users to interact with the data and gain a deeper understanding of app performance. Overall, this project highlights the importance of data analytics in making informed business and development decisions.